Tribhuvan University

Faculty of Humanities & Social Sciences

OFFICE OF THE DEAN

Network and Data Communications

BCA — Semester 4 — — Answer Sheet

Subject Code:

Note: This answer sheet is currently a template. Detailed answers will be added progressively. Students are encouraged to refer to textbooks and class notes.

Group B

1. Explain the concept of Binary Amplitude Shift Keying. Represent bit sequence 110010110 by the following waveforma. Manchesterb. Differential Manchester

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

2. Explain the design of Stop-and-Wait ARQ. Illustrate it with suitable flow diagram example.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

3. What are the features of Link State routing protocols. Explain how Link State routing protocol can be used to find the shortest path with relevant example. What are its disadvantages.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

4. Differentiate between switch, router, and hub.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

5. What are the major differences between noise, distortion and attenuation?

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

6. A slotted ALOHA network transmits 200-bit frames using a shared channel with a 200-kbps bandwidth. Find the throughput if the system considering all stations together produces 250 frames per second. What do you mean by vulnerable time of slotted ALOHA?

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

7. Why do we use NAT? Explain its different types.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

8. Describe Quality of Service. What are its practical significances?

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

9. What is FTP and how does it work?

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

10. Explain Time Division Multiplexing with required figure.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

11. What is MAC-address? The message sequence is 1101011011 and generator polynomial G(X) = x4+x+1. Calculate the transmitted encoded frame.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

12. Write short notes on:a. Open-loopb. POP

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

These answer sheets are prepared for educational purposes. Students are encouraged to verify with official textbooks and course materials.